

Curriculum Vitae

Mariano Franco Hurtado

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Professional Profile

Civil Engineer graduated from Tecnológico de Monterrey with experience in **construction management**, Building Information Modeling (**BIM**), topographic surveying, Ground Penetrating Radar (**GPR**), and the application of innovative technologies such as **Artificial Intelligence**. Focused on developing practical solutions that reduce processing time and improve accuracy, efficiency, and collaboration in engineering and construction projects.

Education

Tecnológico de Monterrey, Guadalajara Campus

B.S. in Civil Engineering | Aug. 2021 – Jun. 2025

University of Castilla-La Mancha (UCLM), Ciudad Real, Spain

International Exchange Program | Aug. 2024 – Jan. 2025

Professional Experience

Tecno Terra, Georadar y Polímeros S.A. de C.V. — Surveying Manager & Operations Engineer

Guadalajara, Jalisco, Mexico

Oct. 2025 – Present

Internal promotion: Operations Engineer, Oct. 2025 – Jan. 2026; Surveying Manager & Operations Engineer, Jan. 2026 – Present.

- Contributed to establishing and standardizing the Surveying department by defining workflows, quality criteria, delivery procedures, and coordinating georeferenced topographic surveys.
- Conducted and interpreted Ground Penetrating Radar (GPR) surveys to identify potential buried utilities, subsurface anomalies, and ground conditions across projects in multiple regions of Mexico.
- Integrated GPR results with coordinates and georeferenced drawings to produce technical deliverables supporting engineering, excavation, and construction decisions.
- Developed Excel VBA/macros to structure and process coordinate data, survey lines, and field records.
- Implemented AutoHotkey and AutoCAD automations to streamline labeling, line representation, file organization, and drawing preparation.
- Reduced manual information-processing time by 70% per project, improving deliverable consistency and reducing the risk of transcription errors.

AIO Constructora — Junior Engineer | Structural Design & BIM

Zapopan, Jalisco, Mexico

Feb. 2025 – Jun. 2025

- Supported the structural design and construction documentation of an industrial warehouse, including drawings, construction details, technical specifications, and engineering criteria.
- Developed structural analyses and technical calculation reports to validate structural steel and/or reinforced concrete elements according to project requirements.
- Produced and coordinated structural, architectural, and MEP drawings through BIM methodology.
- Collaborated in the preparation of construction documentation, including technical specifications, construction details, quantity takeoffs, and constructability reviews.
- Used Revit, AutoCAD, Navisworks, and applicable design criteria and standards, including AISC, NTC, and local building regulations.

Selected Projects

Circular Economy Project for Diageo

Jalisco, Mexico Jan. 2025 – May 2025
Developed a proposal to recover and revalue glass within the tequila industry, aiming to reduce environmental impact and support the company’s sustainability objectives.

Jalisco Logistics Center (CLJ) Project

Jalisco, Mexico Jan. 2024 – Jun. 2024
Developed a proposal for a multimodal airfield at the Jalisco Logistics Center, integrating an airstrip and rail connections.

CONAGUA Project

Guadalajara, Jalisco, Mexico Aug. 2023 – Dec. 2023
Designed a drinking-water supply and wastewater network for a community of 223 residents in Mezquitic, Jalisco.

Technical Skills

AutoCAD • CivilCAD • QGIS • EPANET • Revit • SketchUp • STAAD.Pro • RAM Elements • RAM Connection • Navisworks • Excel VBA & Macros • Power BI • Leica Infinity • GPR-Slice • AutoHotkey • Technical Drawings • Generative AI Applied to Engineering • Technical Prompt Engineering

Professional Skills

Leadership • Problem Solving • Critical Thinking • Effective Communication • Teamwork • Attention to Detail • Adaptability • Creativity & Innovation • Mentoring • Personnel Management • Ethical Responsibility • Empathy • Proactivity

Languages

Spanish: Native
English: Upper-Intermediate (IELTS B2, Feb. 2024)

Certifications

Microcredential in Introduction to the Digital Era and Big Data

— Tecnológico de Monterrey, 2025

Microcredential in Fundamentals of Data Analysis with Power BI

— Tecnológico de Monterrey, 2025